Homework 5

Exercise List

Undergrad CMOS Course – 2026
Fundación FULGOR
Professors: Ramón Sarmiento and Camilo Estévez

General Assumptions

Unless otherwise stated, use the following parameters for all exercises:

$$\mu_n C_{\text{ox}} = 200\,\mu\text{A/V}^2, \qquad \mu_p C_{\text{ox}} = 100\,\mu\text{A/V}^2, \qquad \lambda = 0,$$
$$V_{TH,n} = 0.4\,\text{V}, \qquad V_{TH,p} = -0.4\,\text{V}.$$

Part I — Exercises

Exercise 1 — Problem 7.2

Design the circuit shown in Fig. 7.40 for a drain current of 1 mA. If $W/L = 20/0.18$, calculate R_1 and R_2 such that the input impedance is at least 20 kΩ.

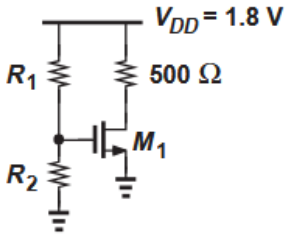


Figure 7.40

Exercise 2 — Problem 7.4

The circuit shown in Fig. 7.42 must be designed for a voltage drop of 200 mV across R_S.

(a) Calculate the minimum allowable value of W/L if M_1 must remain in saturation.

(b) Calculate the required values of R_1 and R_2 if the input impedance must be at least 30 kΩ.

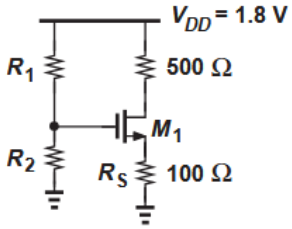


Figure 7.42

Exercise 3 — Problem 7.6

The self-biased stage shown in Fig. 7.44 must be designed for a drain current of 1 mA. If M_1 must provide a transconductance of $1/(100\,\Omega)$, calculate the required value of R_D. Neglect R_G.

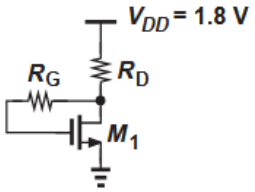


Figure 7.44

Exercise 4 — Problem 7.17

In the common-source stage shown in Fig. 7.54, $W/L = 30/0.18$ and $\lambda = 0$.

(a) What gate voltage yields a drain current of 0.5 mA? Verify that M_1 operates in saturation.

(b) With such a drain bias current, calculate the voltage gain of the stage.

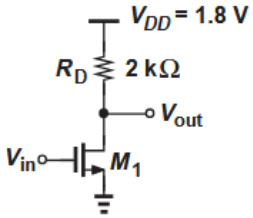


Figure 7.54

Exercise 5 — Problem 7.20

The common-source stage shown in Fig. 7.56 must provide a voltage gain of 10 with a bias current of 0.5 mA. Assume $\lambda_1 = 0.1\,\mathrm{V}^{-1}$ and $\lambda_2 = 0.15\,\mathrm{V}^{-1}$.

(a) Calculate the required value of $(W/L)_1$.

(b) If $(W/L)_2 = 20/0.18$, calculate the required value of V_B.

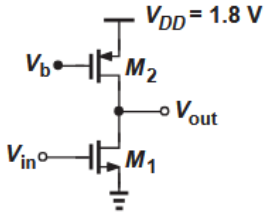


Figure 7.56

Exercise 6 — Problem 7.25

Design the circuit shown in Fig. 7.59 for a voltage gain of 3. If $(W/L)_1 = 20/0.18$, determine $(W/L)_2$. Assume $\lambda = 0$.

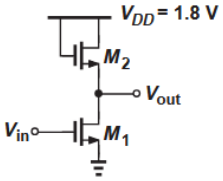


Figure 7.59

Part II — Video Assignment

First, watch the tools installation video available in Classroom. Then, follow the tutorial in the playlist below through Part 4, *Netlist Generation*:

YouTube playlist